

Self

Reducing Runtime Overhead (JITting):

- **Problem:** JIT compiling code at execution time introduces startup latency and CPU overhead.
- **Solutions & Techniques:**
 - **Ahead-Of-Time (AOT) Compilation / Native AOT:** Compiles CIL directly into native code before execution, bypassing the JIT compiler completely at startup.
 - **Tiered Compilation:** Starts with unoptimized fast-compiled code (Tier 0) for quick app startup, then re-compiles frequently executed methods ("hot code") into highly optimized native code (Tier 1) in the background.
 - **ReadyToRun (R2R) Images:** Pre-compiles parts of the assembly to native code ahead of time while maintaining CIL fallbacks, significantly improving startup time.
 - **Profile-Guided Optimization (PGO):** Collects data about how the application runs to allow the JIT compiler to optimize code paths based on real usage patterns.